

11. Symbol Table Operations

Operations: **Insert, Search, Display, Delete**

```
#include <stdio.h>

for (i = 0; i < n; i++)
{
    if (strcmp(table[i].name, name) == 0)
    {
        printf("Symbol Found\n");
        printf("Name: %s\n", table[i].name);
        printf("Type: %s\n", table[i].type);
        printf("Address: %d\n", table[i].address);
        return;
    }
}

printf("Symbol not found.\n");
}

void display()
{
    int i;

    printf("\nSymbol Table\n");
    printf("Name\tType\tAddress\n");

    for (i = 0; i < n; i++)
        printf("%s\t%s\t%d\n",
            table[i].name,
```

```
        table[i].type,  
        table[i].address);  
}
```

```
void deleteSymbol()  
{  
    char name[30];  
    int i, j;  
  
    printf("Enter symbol to delete: ");  
    scanf("%s", name);  
  
    for (i = 0; i < n; i++)  
    {  
        if (strcmp(table[i].name, name) == 0)  
        {  
            for (j = i; j < n - 1; j++)  
                table[j] = table[j + 1];  
  
            n--;  
            printf("Symbol deleted.\n");  
            return;  
        }  
    }  
  
    printf("Symbol not found.\n");  
}
```

```
int main()
{
    int choice;

    do
    {
        printf("\n1. Insert");
        printf("\n2. Search");
        printf("\n3. Display");
        printf("\n4. Delete");
        printf("\n5. Exit");
        printf("\nEnter choice: ");
        scanf("%d", &choice);

        switch (choice)
        {
            case 1: insert(); break;
            case 2: search(); break;
            case 3: display(); break;
            case 4: deleteSymbol(); break;
            case 5: printf("Exiting...\n"); break;
            default: printf("Invalid choice.\n");
        }
    }
    while (choice != 5);

    return 0;
}
```

OUTPUT

```
1. Insert
2. Search
3. Display
4. Delete
5. Exit
Enter choice: 1
Enter symbol name: a
Enter type: int
Enter address: 100
Symbol inserted successfully.
```

```
1. Insert
2. Search
3. Display
4. Delete
5. Exit
Enter choice: 2
Enter symbol to search: b
Symbol not found.
```

```
1. Insert
2. Search
3. Display
4. Delete
5. Exit
Enter choice: 3
```

```
Symbol Table
Name    Type    Address
a       int     100
```